RR-G-661E
21 March 1984
SUPERSEDING
RR-G-661D
July 20, 1978

FEDERAL SPECIFICATION

GRATING, METAL, BAR TYPE (FLOOR, EXCEPT FOR NAVAL VESSELS)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bar-type floor gratings.

1.2 Classification. The grating shall be of the following one types, classes, and composition as specified (see 6.2).

Type I - Parallel bearing bars, with right angle cross members.

Class 1 - Welded (steel or stainless steel)
Class 2 - Pressure locked (steel, stainless steel, or aluminum)

Type II - Parallel bearing bars, with riveted diagonal cross members.

1.2.1 Composition.

Material A - Aluminum
Material S - Steel
Material SS - Stainless steel

1.2.2 Surfaces. Unless specified as serrated (see 6.2), the surface of types I and II shall be plain.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-S-781 - Strapping, Steel, and Seals

Federal Standard:

Fed-Std-123 - Marking for Shipment (Civil Agencies)

FSC 5670

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(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing of

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM):

A123 - Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars, and Strip
A153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A167 - Stainless and Heat-resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A569 - Steel Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality
A659 - Steel Carbon (0.16 Maximum to 0.25 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality
B210 - Aluminum-Alloy, Drawn Seamless Tubes
B221 - Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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National Association of Architectural Metal Manufacturers (NAAMM):

Metal Bar Grating Manual

(Application for copies should be addressed to the National Association of Architectural Metal Manufacturers, 100 South Marion Street, Oak Park, IL 60302.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish metal bar grating for first article inspection and approval (see 4.2.1 and 6.2).

3.2 Standard commercial product. The metal bar grating shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the metal bar grating being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.3.1 Aluminum. Aluminum extrusions shall conform to ASTM B221 and drawn aluminum shall conform to ASTM B210, unless otherwise specified (see 6.2). The aluminum gratings will be supplied in the standard mill finish as fabricated.

3.3.2 Steel. Steel shall be weld quality mild carbon steel conforming to ASTM A569 or ASTM A659. Unless specified (see 6.2), steel grating shall be zinc coated in accordance with ASTM A123 after fabrication. Steel hardware and steel fasteners shall be zinc coated in accordance with ASTM A153, unless otherwise specified (see 6.2).

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3.3.3 Stainless steel. Stainless steel shall conform to ASTM A167, types 302 or 304.

3.4 Design and construction. The metal bar grating shall be designed and constructed to the size, load and deflection requirements, and tolerances of the Metal Bar Grating Manual (see 6.2).

3.4.1 End-banding bars. When specified (see 6.2), end-banding bars shall be included on type I and II gratings and shall be the same size as the bearing bars.

3.5 Workmanship.

3.5.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.5.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.5.3 Zinc coating. Steel grating shall be zinc coated in accordance with ASTM A-123 after fabrication.

3.5.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.6 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one grating of the type, class, material and size specified when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the preparation for delivery inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All gratings of the same type, class, material, and size offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level (AQL) shall be of 1.5 percent defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each metal bar grating selected shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement shall constitute a defect.

4.5 Preparation for delivery inspection. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one unit prepared for shipment. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. Packing shall be in accordance with MIL-STD-794. Contents of containers shall not exceed 4,000 pounds.

5.1.2 Level B. Gratings of like description shall be bundled in secured lifts not to exceed 4,000 pounds. The gratings shall be arranged and secured with steel strapping conforming to QQ-S-781, class 1, type I, finish B, size 0.035 by 1-1/4 inch. Nominal 2- by 4-inch wood battens shall be placed under each strap on top, bottom, and sides. The strapping shall be stapled to the wood battens. A minimum of three straps shall be provided for lengths up to 12 feet. The center strap shall be spaced not to exceed one (1) inch from the center of the length of the lift. The end straps shall be placed at the same distance (± 1 inch) from either end of a lift, shall be not nore than 18 inches from each end of the lift, and shall be at the same distance from the ends for all lifts of the same length. For longer lengths, add one intermediate strap for each additional 6 feet or fraction thereof and equally space all intermediate straps.

5.1.3 Commercial. The gratings shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.1.3.1 Unitization. When specified (see 6.2), carload or truckload shipments shall be unitized in accordance with the contractor's standard practice. The unitized load shall not exceed 2,500 pounds in weight and 63 inches in height.

5.2 Marking.

5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The gratings covered by this specification are intended to cover trenches, drains, loading ramps, catwalks, stair treads, and platforms, and in certain areas, to be used as flooring.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, class, and material required (see 1.2).

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c. When serrated surface is required (see 1.2.2).
d. When a first article is required (see 3.1).
e. Finish of gratings and fasteners other than specified (see 3.3.1 and 3.3.2).
f. Size of grating required (see 3.5).
g. When end banding bars shall be included (see 3.5.1).
h. Level of packing required (see 5.1).
i. When unitization shall be in accordance with 5.1.3.1 is required (see 5.1.3.1).

MILITARY INTERESTS:

Custodians

Navy - YD
Air Force - 99

Review

Air Force - 84

User activity

Army - ME

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSC

PREPARING ACTIVITY:

Navy - YD

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